

PS-MPs promotes the progression of inflammation and fibrosis in diabetic nephropathy through NLRP3/Caspase-1 and TGF- β 1/Smad2/3 signaling pathways

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ABSTRACT

Background: Diabetic nephropathy (DN) is a prevalent chronic microvascular complication of diabetes and the leading cause of end-stage renal disease (ESRD). Understanding the progressive etiology of DN is critical for the development of effective health policies and interventions. Recent research indicated that polystyrene microplastics (PS-MPs) contaminate our diets and accumulate in various organs, including the liver, kidneys, and muscles.

Methods: In this study, ten-week-old db/db mice and db/m mice were fed. Besides, db/db mice were divided into two groups: PS-MPs group (oral administration of 0.5 μ m PS-MPs) and an H₂O group, and they were fed for three months. A type II diabetes model was established using db/db mice to investigate the effects of PS-MPs on body weight, blood glucose level, renal function, and renal fibrosis.

Results: The results demonstrated that PS-MPs significantly exacerbated various biochemical indicators of renal tissue damage, including fasting blood glucose, serum creatinine, blood urea nitrogen, and blood uric acid. Additionally, PS-MPs worsened the pathological alterations and degree of fibrosis in renal tissue. An increased oxidative stress state and elevated levels of inflammatory cytokines, such as tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and monocyte chemoattractant protein-1 (MCP-1) were identified. Furthermore, PS-MPs significantly enhanced renal fibrosis by inhibiting the transition from epithelial cells to mesenchymal cells, specifically through the inhibition of the TGF- β /Smad signaling pathway. The expression levels of NOD-like receptor protein 3 (NLRP3), apoptosis-associated speck-like protein containing a CARD (ASC), Caspase-1, and cleaved Caspase-1, which are inflammasome proteins, were significantly elevated in the PS-MPs group.

Conclusion: The findings suggested that PS-MPs could aggravate kidney injury and renal fibrosis in db/db mice by promoting NLRP3/Caspase-1 and TGF- β 1/Smads signaling pathways. These findings had implications for elucidating the role of PS-MPs in DN progression, underscoring the necessity for additional research and public health interventions.

Abbreviations: DN, Diabetic nephropathy; ESRD, End-stage renal disease; PS-MP, polystyrene microplastics; MPs, Microplastics; TNF- α , Tumor necrosis factor- α ; MCP-1, Interleukin-1 β ; IL-1 β Monocyte chemoattractant protein-1; NLRP3, NOD-like receptors family pyrin domain containing 3; ASC, Apoptosis-associated speck-like protein containing a CARD; DKD, Diabetic kidney disease; EMT, Epithelial-mesenchymal transition; ECM, Extracellular matrix; TGF- β 1, Transforming growth factor- β 1; Col-1, Collagen I; FN, Fibronectin; α -SMA, α -smooth muscle actin.

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1. Introduction

Since the 1950s, plastics have been molded into a diverse array of products, resulting in their growing prevalence owing to their convenience and affordability (Geyer et al., 2017; Rhodes, 2018). Global plastic production reached 355 million tons in 2016, and it is projected that plastic production will reach 33 billion tons by 2050 (Alimba and Faggio, 2019; Rochman et al., 2013). However, the ineffective rates of plastic recycling and biodegradation have resulted in a significant accumulation of plastic products in the environment (Díaz-Torres et al., 2017; Jambeck et al., 2015). Plastic products mainly contain harmful substances, such as di(ethylhexyl) phthalate, bisphenol A, and bisphenol S, which pose risks to human health (Shi et al., 2021; Zhang et al., 2021). The toxic effects of microplastics (MPs) have remarkably attracted scholars' attention (Wan et al., 2020). Prior research suggested that MPs can be accumulated in the human body through the food chain (Smith et al., 2018). Furthermore, emerging evidence indicated that MPs can be accumulated in the gastrointestinal tract, liver, and kidneys of mice (Deng et al., 2017a). Among various types of MPs found in the ocean, polystyrene is one of the most common polymers (Ross et al., 2021). Exposure to polystyrene microplastics (PS-MPs) has been demonstrated to induce damage to various organs, including ovaries (Wan et al., 2020), lungs (Dong et al., 2020), liver (Feng et al., 2021), and kidneys (Wang et al., 2021).

Diabetic nephropathy (DN) is a prevalent chronic microvascular complication of diabetes, and it affects a significant proportion of diabetic patients, with approximately 40% eventually developing diabetic kidney disease (DKD) or progressing to end-stage renal disease (ESRD) (Yaribeygi et al., 2019). It is estimated that the number of diabetic patients will globally reach 700 million by 2045 (Saeedi et al., 2019). DN is a consequential complication of diabetes and its incidence rises in parallel with the increasing prevalence of diabetes in the population (Zeng et al., 2014). However, it remains elusive whether PS-MPs are a contributing risk factor for DN.

Tubulointerstitial fibrosis is an essential pathological change in the development of DN (Wei et al., 2013; Loeffler and Wolf, 2015). The most common biological process in renal fibrosis is epithelial-mesenchymal transition (EMT), which is characterized by the loss of epithelial phenotype and acquisition of pro-fibrotic features (Wei et al., 2013). When EMT is activated, renal fibroblasts transform into myofibroblasts, leading to the secretion of excessive extracellular matrix (ECM), and ultimately resulting in fibrosis (Loeffler and Wolf, 2015). Studies have demonstrated that transforming growth factor- β 1 (TGF- β 1) can induce EMT activation in renal tubular epithelial cells (Lan, 2012; Wang et al., 2017). TGF- β 1 phosphorylates downstream proteins Smad2 and Smad3, increasing the expression level of α -smooth muscle actin (α -SMA) (Lu et al., 2019). Therefore, it is necessary to investigate whether PS-MPs can promote EMT through the TGF- β 1/Smad signaling pathway in DN.

Numerous studies have reported a strong association between an inflammatory response and the occurrence and progression of DN (Fernandes et al., 2016; Donate-Correa et al., 2020). The activation of NOD-like receptor protein 3 (NLRP3) inflammasomes has exhibited to stimulate the secretion of proinflammatory cytokines (e.g., interleukin-18 (IL-18) and IL-1 β), exacerbating renal inflammation and fibrosis (Zhu et al., 2020). In a mouse model of DN, the knockout of NLRP3 has significantly attenuated the renal inflammatory response and improved renal function (Wu et al., 2018).

The present research aimed to elucidate the role and underlying mechanism of oral exposure to PS-MPs in the development of DN. The findings indicated that PS-MPs could lead to adverse outcomes, including impaired kidney function, fibrosis, oxidative stress, and inflammation in the renal tissue of db/db mice. Furthermore, it was attempted to investigate the specific mechanism by which PS-MPs could promote fibrosis in mice. The results revealed that PS-MPs could activate the TGF- β 1/Smads signaling pathway, while the NLRP3/Caspase-1 signaling pathway could exacerbate renal inflammation in mice. These

findings may provide valuable insights for future studies aiming to comprehend the mechanisms underlying PS-MPs-induced injury in DN.

2. Materials and methods

2.1. Characterization of MPs

The Baseline Chromtech Research Center (Tianjin, China) provided PS-MPs that had a diameter of 0.5 μ m. A suspension of MP beads was made in distilled water with a final concentration of 1 mg/L. The pH of the water was measured to be 6.9 ± 0.1 . PS-MPs were determined by scanning electron microscopy (SEM) (Quanta-250, Hillsboro, USA). The zeta potential of 0.5 μ m PS-MPs was -9.172 ± 0.426 . The working suspensions were made by diluting the stock solution with liquid K-medium. Before exposure, a 30-min sonication at 40 kHz (100 W) was conducted to ensure that the working suspensions achieved monodispersity.

2.2. Animal experiments

The animal experiments conducted in this study adhered to the guidelines outlined in the National Institutes of Health Guide for the Care and Use of Laboratory Animals. The research protocol was reviewed and approved by the Institutional Animal Care and Use Committee (IACUC) of the Third Hospital of Wuhan University (Wuhan, China; Reference No. SY2022-015). Mice were housed under a 12-h light/dark cycle, and high-quality bedding was provided that was odorless, non-toxic, sterile, and absorbent. The bedding was regularly changed, and mice had unrestricted access to standard rodent diet and water.

Recent research has demonstrated striking similarities in the disease progression between db/db mice and patients with type-2 diabetes (T2D). Consequently, db/db mice have emerged as an optimal animal model for investigating the pathogenesis and treatment strategies of T2D (Sharma et al., 2003). In the present study, 10 ten-week-old male db/m mice and 20 male db/db mice were purchased from Three Gorges University (Yichang, China). All mice were maintained in a specific-pathogen-free (SPF) environment at a temperature of 22–23 °C. Following one-week acclimation, mice were subsequently partitioned into three distinct groups, each consisting of 10 mice. Exposure doses were calculated based on concentrations of PS-MPs in the aquatic environment and recent toxicology studies (Shan et al., 2022). The control group included db/m and db/db mice that received H₂O for 12 weeks. PS-MPs were administered to db/db mice for 3 months that were allocated to the experimental group. After 3 months, mice were anesthetized with 2% pentobarbital sodium (30 mg/kg), and upon achieving a state of anesthesia, euthanasia was carried out via cervical dislocation. Thereafter, kidney tissue was collected, and blood sample was tested to detect biochemical indices. The biochemical indicators included serum creatinine, blood urea nitrogen, and blood uric acid. The fluorescent MP particles in the wavelength range of 620/680 (red) were purchased from the Baseline Chromtech Research Center, which dissolved in pure water to prepare a mixed solution of 100 mg/L and then fed to mice for 7 days (0.1 ml/d). Mice were euthanized, and their kidney tissue was removed for frozen sectioning (GP1008-1; Servicebio, Wuhan, China). Fluorescent MP particles were subsequently observed under a microscope (NIKON DS-U3).

2.3. Sample collection and immunohistochemical analysis

Mice were subjected to an overnight fasting period, following which sodium pentobarbital (150 mg/kg) was administered to induce anesthesia. Thereafter, an evaluation was conducted on the body weight, liver weight, kidney weight, and fasting blood glucose (FBG) level of each individual mouse. Upon completion of the trial, FBG level was measured by a blood glucose meter subsequent to a 12-h period of

refraining from food consumption. In order to extract serum, blood samples were collected from each animal and were subsequently separated by centrifugation at 1200 g for 15 min at a temperature of 4 °C. Serum samples were collected to measure the concentrations of serum creatinine (SCr), blood urea nitrogen (BUN), and blood uric acid (BUA). The kidney tissue was obtained and partitioned along the sagittal plane into two portions. One portion was subjected to fixation in a 10% formalin solution, while the other portion was preserved at a temperature of −80 °C.

2.4. Biochemical analysis

In this study, SCr level was quantified using a commercially available creatinine detection kit (S03076, Shenzhen Rayto Biological Co., Ltd., Shenzhen, China). The concentrations of BUN (S03036, Shenzhen Rayto Biological Co., Ltd.) and uric acid (UA; S03035, Shenzhen Rayto Biological Co., Ltd.) in serum were determined using a biochemical kit (Shenzhen Rayto Biological Co., Ltd.). Blood samples were obtained from murine eyeballs and subjected to centrifugation at 3000–4000 rpm for 10 min using a tabletop high-speed centrifuge machine (D3024R, DLAB, Beijing, China). The resulting supernatant was carefully collected and processed according to the instructions provided with the respective kits. Subsequently, the prepared samples were subjected to analysis using an automated biochemical analyzer (Chemray 800, Shenzhen Rayto Biological Co., Ltd.). Specifically, the enzymatic method was employed for SCr measurement, the urease-glutamate dehydrogenase method was utilized for BUN detection, and the uricase method was applied for UA detection. All experimental procedures were faithfully carried out in compliance with the manufacturer's guidelines.

2.5. Histological evaluation

The kidney tissue was stored in a solution of formalin with a concentration of 10%. It was further fixed using paraformaldehyde with a concentration of 4%. The tissue was then dried, embedded into paraffin, and finally cut into sections with a thickness of 4 µm. Following the solidification of the mounting medium, the sections were subjected to scanning using the Nikon Imaging system, specifically the NIKON DS-U3 (Nikon, Tokyo, Japan). The sections were subjected to staining using the hematoxylin-eosin (H&E), periodic acid-Schiff (PAS), and Masson's trichrome (Masson) procedures. A light microscope (Eclipse TE2000-U, Nikon) was utilized to observe the morphology of kidney tissue, followed by histopathological analysis. The results of H&E staining were reported descriptively. The degree of tubulointerstitial injury was assessed using ImageJ software by evaluating 10 fields of view per section. Renal glomeruli and interstitial tubules were observed at a magnification of ×40. The extent of tubulointerstitial injury was assessed by calculating the area ratio of tubular dilation/flattening, loss of brush border, tubular cyst formation, tubular atrophy, and vacuolization. Thereafter, 10 randomly selected overlapping areas (200× magnification) were scored for tubular necrosis, tubular formation, tubular dilation, and loss of brush border. The scoring system was summarized as follows: 0 = absent, 1 = ≤10%, 2 = 11–25%, 3 = 26–45%, 4 = 46–75%, 5 = ≥76% (Kuang et al., 2023; Xiong et al., 2023). The assessment of glomerular damage was conducted through the application of PAS staining technique, followed by the quantification of damage via computing the average optical density (AOD). The glomerular sclerosis index was obtained by multiplying the severity of the lesion (0–4+) by the percentage of glomeruli affected by the same severity, reflecting the increase in mesangial matrix or glomerular sclerosis. The damage extent of each individual tissue sample was subsequently determined by summing up these scores. (a 1+ lesion involved 25% of glomeruli, while a 4+ lesion involved 100% of glomeruli.) (Raij et al., 1984).

The evaluation of tubulointerstitial fibrosis, using Masson staining, involved the computation of the proportion of collagen area. This was

achieved by dividing the collagen area by the sum of the positive area and the total area, followed by multiplying the result by 100.

2.6. Immunohistochemistry (IHC)

The segments were baked for 1 h at 60 °C. Each section was rinsed twice with xylene (Sigma-Aldrich, MO, USA), twice with 100% ethanol (Sigma-Aldrich) for 10 min, once with 95% ethanol for 5 min. and once with 75% ethanol. Sections were simmered in citrate buffer [0.01 M (pH 6.0) with 1% Tween 20] for 30 min prior to immunostaining. IHC was conducted in accordance with the established protocol. In this experiment, the sections were treated with hydrogen peroxide (#TA-060-H2O2Q; Thermo Fisher Scientific, Waltham, MA, USA) for 10 min to induce blocking. Following this, the sections were subjected to thrice washing with TBS-T for 15 min each. Subsequently, the sections were blocked with UltraVision protein blocking agent (TL-060-QHL; Thermo Fisher Scientific) at room temperature for 10 min.

The slides were incubated with antibodies targeting F4+/80 (1: 1000, GB113373–100, Servicebio), IL-1β (1: 800, GB11113–100, Servicebio), MCP-1 (1: 1000, GB11199–100 Servicebio), fibronectin (1:500; #nb100–2324; Novusbio, Centennial, CO, USA), α-SMA (1:2000; #67735–1-Ig; Proteintech, Rosemont, IL, USA), type I collagen (Col-I) (1:1000; #A7341; ABclonal Technology, Woburn, MA, USA), NLRP3 (1:1000; GB114320–100, Servicebio), and Smad2 (1:1000; #A7341; ABclonal Technology) in a solution containing 3% bovine serum albumin (BSA) at 4 °C overnight. Subsequently, the slides were thrice washed with TBS-T for 15 min each at room temperature. The slides were subjected to a 10-min incubation with the primary antibody. Thereafter, they were rinsed with TBS-T for three times, 15 min each time. Afterwards, the slides were treated with horseradish peroxidase (HRP) polymer Quanto (G1213–100UL, Servicebio) for 10 min. Finally, the slides were washed with TBS-T at room temperature for three times, 15 min each time. The slides were subjected to treatment with 3,3-diaminobenzidine (#36201ES03; Veasen), followed by cessation of the reaction using water. Subsequently, the sections were exposed to hematoxylin (#GC307020–5 g, Servicebio) for 3 min at ambient temperature. The kit was utilized in accordance with the manufacturer's instructions for Masson's trichrome staining (#TRM-2-IFU, ScyTek, Logan, UT, USA). Following that, the mounting medium was applied to the slides, and coverslips were installed. Once the mounting medium was solidified, the sections were scanned using the Nikon Imaging system (DS-U3, Nikon). This study included the IHC in 10 fields to indicate the percentage of the positive region.

2.7. Quantitative reverse transcription polymerase chain reaction (RT-qPCR)

The extraction of total RNA from mouse kidney tissue was performed using TRIzol reagent (Invitrogen, Inc., Carlsbad, CA, USA). The assessment of RNA quality was conducted using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). A ratio of absorbance (OD260/280) within the range of 1.9–2.1 indicated satisfactory purity of the RNA. The first-strand cDNA synthesis was performed using the SuperScript First-Strand Synthesis system in accordance with the manufacturer's instructions (Thermo Fisher Scientific). The amplification process was conducted using the SYBR FAST qPCR kit (Servicebio) and the CFX96 real-time PCR system (Bio-Rad, Hercules, CA, USA). Each reaction comprised a 10 µL volume, containing 400 nM concentration of each primer, 1 µL of complementary DNA (cDNA), 5 µL of SYBR Green Master Mix reagent, with the remaining volume supplemented with nuclease-free water. The amplification technique consisted of an initial incubation at 50 °C for 2 min, followed by a denaturation step at 95 °C for 5 min. This was followed by a series of 40 cycles, each consisting of a denaturation step at 94 °C for 15 sec, and an annealing step at 6 °C for 30 sec. All experimental procedures were conducted in triplicate. The relative expression levels were normalized to the reference gene β-actin.

The primer sequences are listed in [Supplementary Table 1](#).

2.8. Western blot analysis

Western blot analysis was conducted following previously established protocols. The kidney tissue was subjected to lysis using radio-immunoprecipitation assay (RIPA) lysis buffer, which was supplemented with 1% phenylmethylsulfonyl fluoride (PMSF) protease inhibitor (#P1005, Beyotime Institute of Biotechnology, Shanghai, China) and phosphatase inhibitor (#P1081, Beyotime Institute of Biotechnology). The extraction of total protein was carried out using a solution consisting of 1% Triton X-100 and 0.1% sodium dodecyl-sulfate (SDS). An equivalent quantity of protein was separated using a 10–15% SDS-polyacrylamide gel electrophoresis (PAGE) gel and subsequently deposited onto polyvinylidene difluoride (PVDF) membranes. After being blocked with a 5% skim milk solution for 1 h, the membranes were incubated overnight at 4 °C with primary antibodies targeting TGF- β 1 (#ab215715), FN (#ab2413), α -SMA (#ab32575), apoptosis-associated speck-like protein containing a CARD domain (ASC) (#ab175449), Caspase-1 (#ab1872), p-Smad2 (#ab53100), Smad2 (#ab40855), p-Smad3 (#ab52903), and Smad3 (#ab40854) obtained from Abcam (Cambridge, UK), as well as cleaved Caspase-1 (#Asp297) and NLRP3 (1:1000; GB114320–100) from Cell Signaling Technology, Inc. (Danvers, MA, USA). The antibodies were diluted using a primary antibody dilution solution (G2025–100 ml, Servicebio).

Following thrice washing with TBS-T, the membranes were subsequently incubated with the appropriate secondary antibodies for 1 h at ambient temperature. The following secondary antibodies were utilized: HRP-conjugated goat anti-mouse IgG (1:3000, GB23301, Servicebio) and HRP-conjugated goat anti-rabbit IgG (1:3000, GB23303, Servicebio). After performing thrice washing with TBS-T, the protein bands were observed by employing an enhanced chemiluminescence solution (#34577, Thermo Fisher Scientific). The β -actin was utilized as a loading control in this study.

2.9. Statistical analysis

The statistical analysis was performed using the GraphPad Prism software (GraphPad Software Inc., San Diego, CA, USA). The experimental data were presented in the form of mean $\pm$ standard error of the mean (SEM). The dissimilarities among groups were evaluated utilizing one-way analysis of variance (ANOVA), followed by Dunnett's test for conducting multiple comparisons. The concept of statistical significance was implemented by defining a threshold of P-value less than 0.05.

3. Results

3.1. The characteristics of PS-MPs

Numerous terrestrial and marine processes, including domestic and industrial wastewater discharges, maritime activities, agricultural runoff, and wastewater treatment plant (WWTP) emissions, contribute to the contamination of aquatic environments by PS-MPs. These MPs were detected in human food and air samples, and individuals with pre-existing health conditions would experience more adverse effects from the ingestion or inhalation of PS-MPs (Liu and You, 2023). To showcase the properties of PS-MPs, SEM was utilized to detect PS-MPs purchased from the Baseline Chromtech Research Center. These PS-MPs had an average diameter of 0.5 μ m. The timeline of mice treated with PS-MPs is displayed in [Supplementary Figure 1 A](#). Under two different magnification ratios, it was observed that the diameter of microplastics was approximating 0.5 μ m ([Supplementary Figures 1B–C](#)). In addition, the Fourier spectral analysis results of these plastics are presented in [Supplementary Figure 1D](#). According to the Fourier spectra, the components of these microplastics are PS-MPs. To further investigate their behaviors, the fluorescent microplastics were administered into mice.

Subsequently, the kidneys of these mice were frozen sectioned and analyzed under a microscope, revealing the presence of these microplastics in the kidney tissue, as depicted in [Supplementary Figures 1E–F](#).

3.2. PS-MPs adversely affected kidney function

The body weight and FBG level of mice in the db/db group and PS-MPs group were significantly higher compared with those in the db/m group, confirming the successful establishment of the T2D model ([Fig. 1A–B](#)). Remarkably, the PS-MPs+db/db group displayed a lower body weight compared with the db/db group ([Fig. 1A](#)). Moreover, exposure to PS-MPs resulted in an elevation of FBG level ([Fig. 1B](#)). FBG is a widely used diagnostic marker for diabetes as it reflects the function of pancreatic β -cells and basal insulin secretion (Dimitriadis et al., 2021). Clinical research has demonstrated that metabolic dysregulation in patients results in hyperglycemia, influencing endothelial cells within renal blood vessels. This phenomenon induces cellular morphological alterations, basement membrane thickening, and vasoconstriction, thereby triggering the onset of DN. Notably, this condition is distinguished by glomerulosclerosis and impaired glomerular filtration function (Gross et al., 2005; Phillips and Molitch, 2002).

Furthermore, kidney function was evaluated by analyzing the levels of BUN and Scr. The levels of these indicators significantly increased in mice treated with PS-MPs ([Fig. 1C–D](#)). Moreover, the administration of PS-MPs into db/db mice led to an increase in blood UA level ([Fig. 1E](#)). These findings confirmed that MPs could worsen the severity of renal damage and diabetes symptoms in db/db mice.

3.3. PS-MPs aggravated histopathological alterations

To elucidate the pathological alterations in renal tissue, H&E, PAS, and Masson's staining methods were employed. In the H&E staining, compared with the db/m group, the db/db group exhibited glomerular hypertrophy, increased thickness of the glomerular basement membrane, and mesangial matrix proliferation. In the db/db group exposed to PS-MPs, these aforementioned renal injury alterations were even more noticeable, which were primarily characterized by a more significant thickening of the glomerular basement membrane and the increased mesangial matrix proliferation. The results of PAS staining further confirmed these alterations, and there was also a noticeable thickening of the renal tubular basement membrane ([Fig. 2A and B](#)).

The glomerular sclerosis index could be examined using the PAS staining (Raij et al., 1984). Remarkably, treatment with PS-MPs significantly exacerbated glomerular injury and renal injury ([Fig. 2D and E](#)). In contrast, Masson's staining facilitated the evaluation of renal fibrosis and exhibited a notable intensification of the blue coloration in the collagen fibers of kidney tissue in db/db mice exposed to PS-MPs, which could be indicative of augmented collagen fiber abundance and deposition. These findings serve as prominent indicators of fibrosis, as illustrated in [Fig. 2C and F](#). Collectively, these findings unequivocally demonstrate that PS-MPs could exacerbate tubulointerstitial injury and renal fibrosis in the db/db group compared with db/m group.

3.4. PS-MPs promoted production of inflammatory factors

To investigate the extent of macrophage infiltration, the levels of F4+/80-positive cells were quantified ([Fig. 3A](#)). The IHC results revealed an increased infiltration of macrophages in the glomeruli and tubulointerstitial regions in the db/db group. Moreover, the exposure to MPs further exacerbated the infiltration of macrophages in the db/db group (brown areas are shown in the F4/80 immune group), demonstrating a more noticeable increase in their quantity. The results revealed a notable infiltration of macrophages following exposure to PS-MPs ([Fig. 3A](#)). MCP-1 and IL-1 β are among the earliest described chemokines and play a crucial role in nephritis (Brüggemann et al., 1989; Tesch, 2008). MCP-1 mediates the release of monocytes from the bone

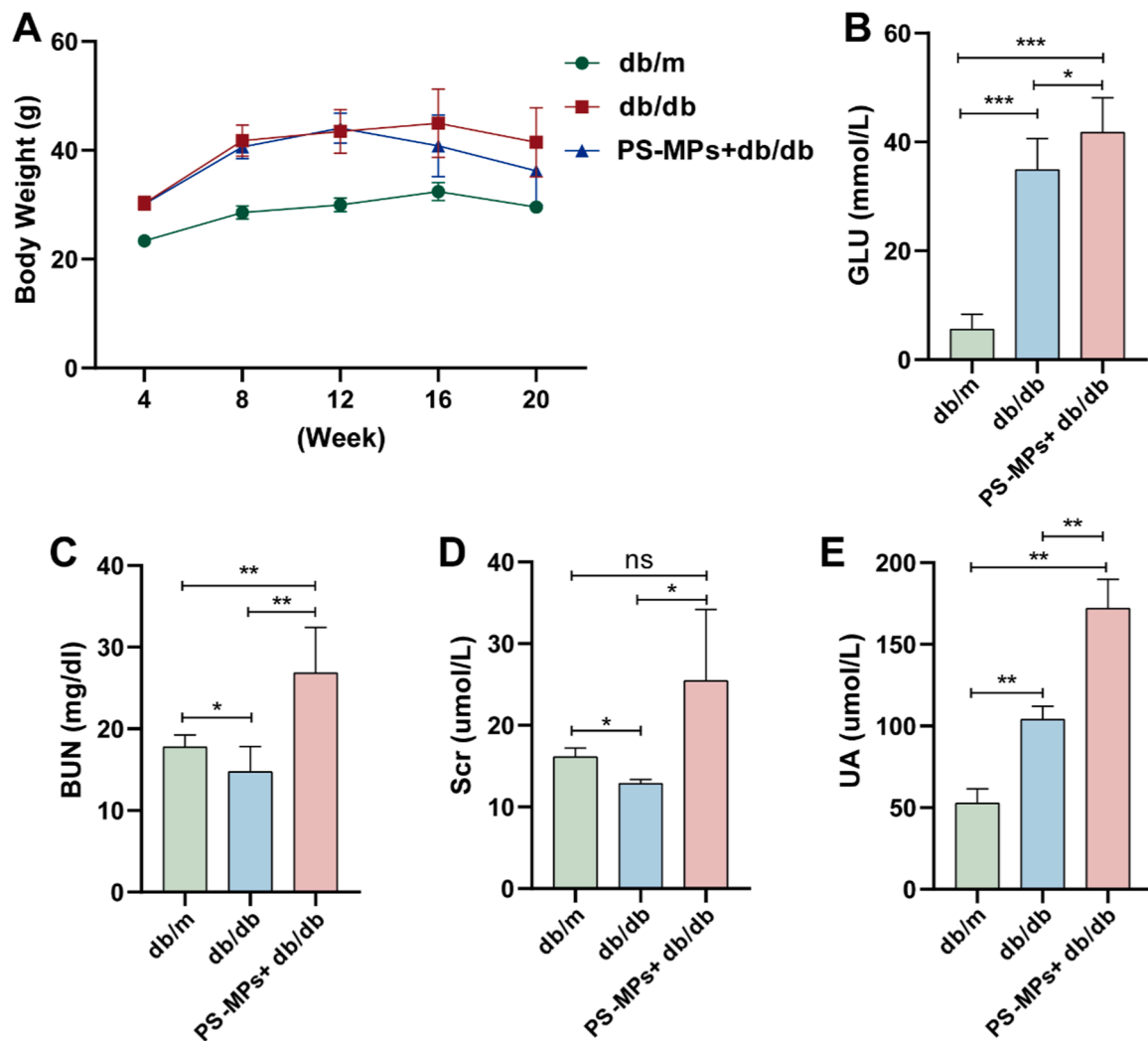


Fig. 1. Effects of PS-MPs on kidney function in db/db mice. (A) the changes of mice in the body weight during the PS-MPs intervention. (B) the level of Fasting-blood glucose (FBG). (C-E) The level of serum blood urea nitrogen (BUN, C), serum creatinine (SCr, D), uric acid (UA, E). Statistical analysis was performed using one-way ANOVA (A-E), $n=10$, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

marrow and then generates a gradient on the endothelial cell glycocalyx, guiding monocytes to the site of inflammation, thereby alleviating the migration of blood leukocytes into inflamed tissue (Haller et al., 2016). IHC results demonstrated a significant upregulation in the protein expression levels of IL-1 β and MCP-1 in the kidneys of db/db mice, which were further augmented by PS-MPs treatment (Fig. 3A-D). To further elucidate the impact of PS-MPs on inflammation in db/db mice, the mRNA expression levels of IL-1 β , MCP-1, IL-6, and TNF- α were assessed (Fig. 3E-H). Remarkably, the mRNA levels of all the four factors in the kidney tissue of db/db mice were significantly elevated after exposure to PS-MPs (Fig. 3E-H). These findings suggested that exposure to PS-MPs could promote macrophage infiltration and upregulate the protein expression levels of the inflammatory factors (IL-1 β and MCP-1).

3.5. PS-MPs increased renal fibrosis in db/db mice by promoting EMT

To investigate the effects of PS-MPs on renal fibrosis, relevant indicators were examined. Notably, fibronectin (FN) is a ubiquitously expressed ECM protein that plays fundamental roles in critical biological processes, such as wound healing, embryonic morphogenesis, and cellular behavior modulation (Zollinger and Smith, 2017). Furthermore, α -SMA stands as the prevailing isoform of actin present in vascular smooth muscle cells, while myofibroblasts, a specialized subtype of

fibroblasts exclusively expressing α -SMA, exert pivotal functions in the pathogenesis of fibrotic reactions (Moreels et al., 2008). In the context of renal fibrosis, an aberrant accumulation and synthesis of Col-I lead to substantial alterations in glomeruli and nephron, thereby impairing normal kidney function (Shintani, 2009). IHC results showed that the protein expression levels of FN, α -SMA, and Col-I were significantly risen in the kidneys of untreated db/db mice, which were further enhanced after exposure to PS-MPs (Fig. 4A-D).

Similarly, the results of Western blot analysis demonstrated that FN and α -SMA expression levels were significantly elevated in the db/db group compared with the control group. Additionally, exposure to PS-MPs further augmented the expression levels of FN and α -SMA (Fig. 4E-G).

3.6. PS-MPs promoted the NLRP3 inflammasome signaling pathway

Zhang et al. pointed out that the NLRP3 inflammasome not only mediates inflammatory responses, but is also associated with pyroptosis, mitochondrial regulation, and myofibroblast differentiation in the renal fibrosis. Inflammation plays a crucial role in the early stages of renal fibrosis, and its activation through the NLRP3/ASC/Caspase-1/IL-1 β axis has a notable function in renal fibrosis (Zhang and Wang, 2019). Therefore, the influences of PS-MPs on the activation of the renal NLRP3

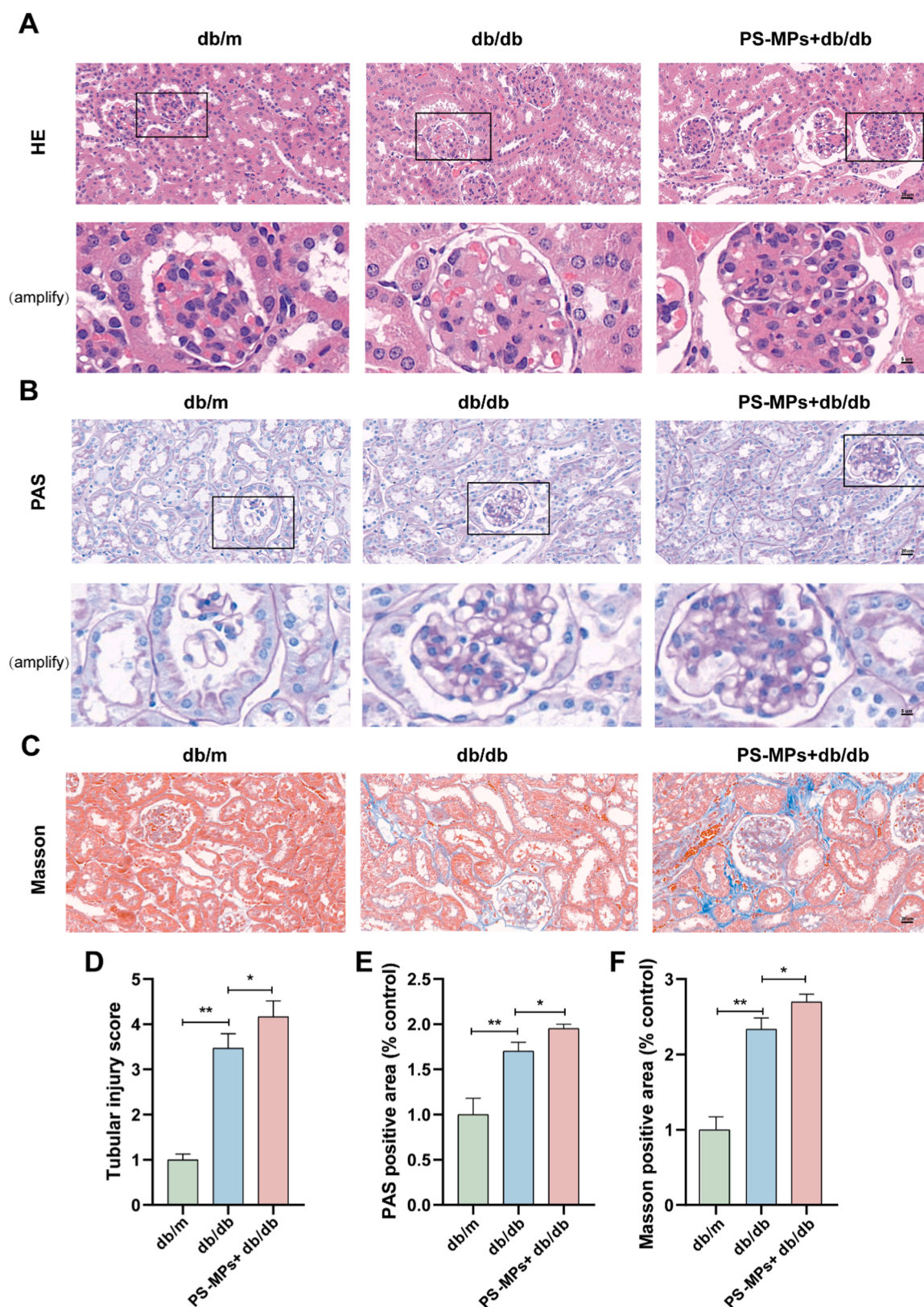


Fig. 2. Effects of PS-MPs on renal pathology in the kidneys of db/db mice. (A-C) Representative images of H&E, PAS, and Masson staining of kidney tissues (400x). Each bar indicates 20 μ m. (D-F) The statistical analysis results of tubular injury score (D), PAS positive area (E) and Masson positive area (F) Statistical analysis was performed using one-way ANOVA (A-D), $n=10$, * $p < 0.05$; ** $p < 0.01$; ns, not significant.

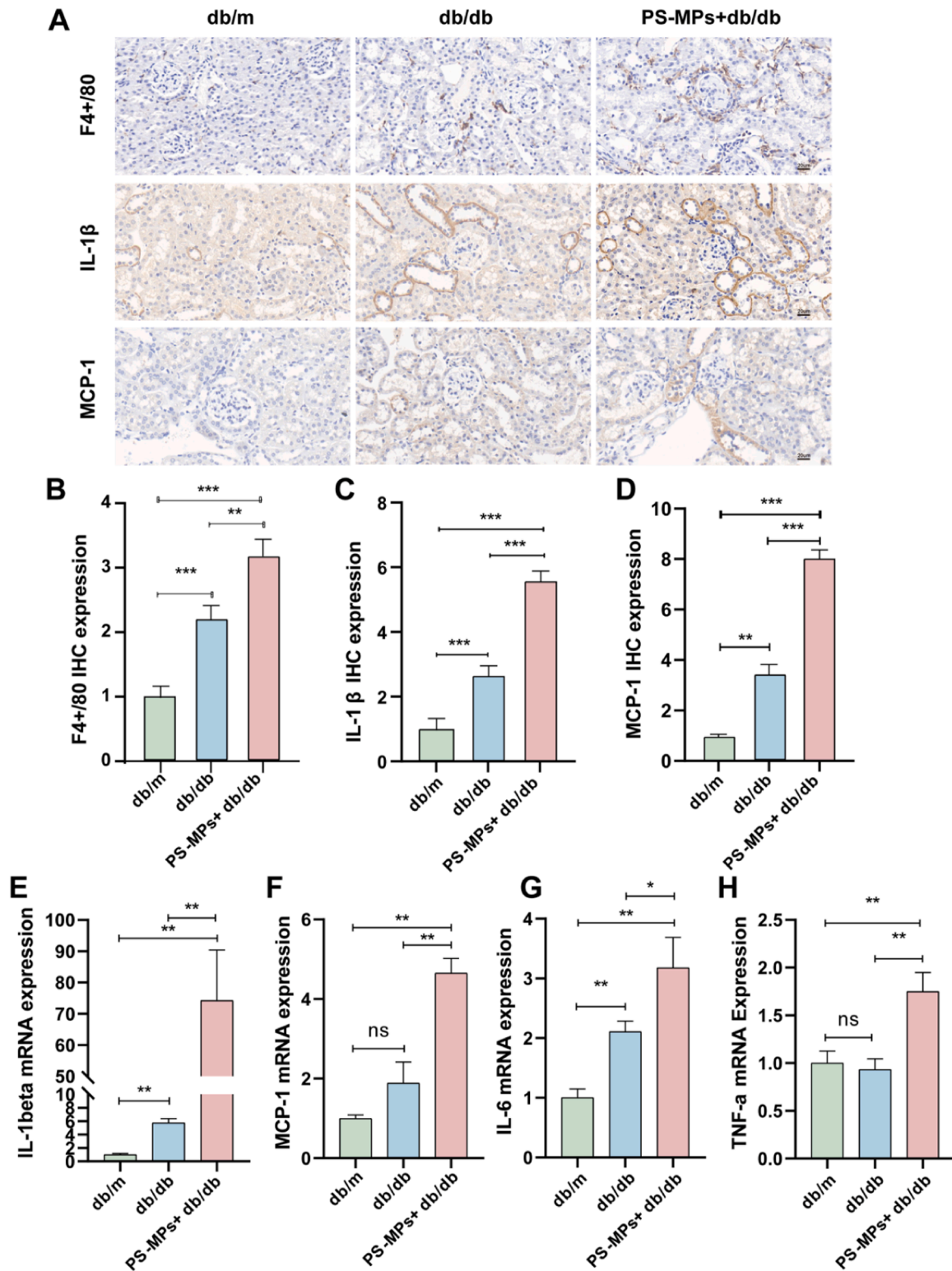


Fig. 3. PS-MPs promote the mRNA production of inflammatory factors. (A-D) Representative immunohistostaining (IHC) micrographs of F4+/80, interleukin-1β (IL-1β), and Monocyte Chemoattractant Protein-1 (MCP-1); magnification, 400×, scale bar, 20 μm. (E-H) mRNA expression level of IL-1β (E), MCP-1 (F), IL-6 (G), and TNF-α (H). Statistical analysis was performed using one-way ANOVA (A-E), n=10, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

inflammasome, a process that has been documented to be crucial in the progression of renal damage, were assessed (Fig. 5A-B). The results of Western blot analysis revealed a notable upregulation in the expression levels of inflammasome proteins, including NLRP3, ASC, Caspase-1, and cleaved Caspase-1, in the renal tissue of mice belonging to the db/db group following exposure to PS-MPs, as compared with the db/db group

(Fig. 5C-H). These findings provided evidence for the potential involvement of PS-MPs in the NLRP3 inflammasome pathway, thereby improving our understanding of the complex mechanisms underlying inflammatory responses.

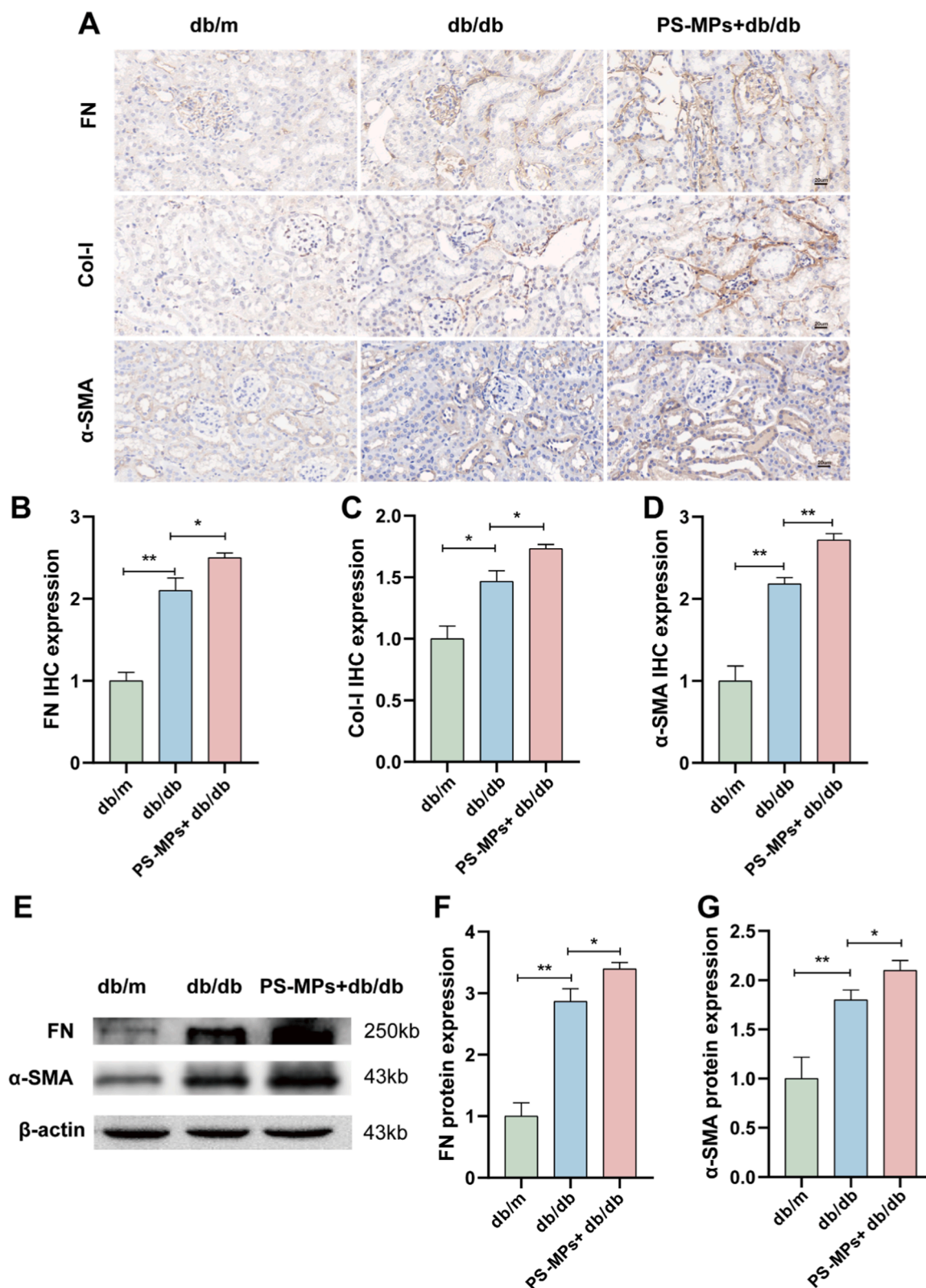


Fig. 4. PS-MPs promote the expression of fibrotic proteins in the kidneys of db/db mice. (A-D) Immunohistochemical results show that MPs promote renal fibrosis in mice. magnification, 400 $\times$, scale bar, 20 μ m. (E-G) Western blot analysis of protein expressions of FN, α -SMA in different groups, Statistical analysis was performed using one-way ANOVA (A-G), n=10, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

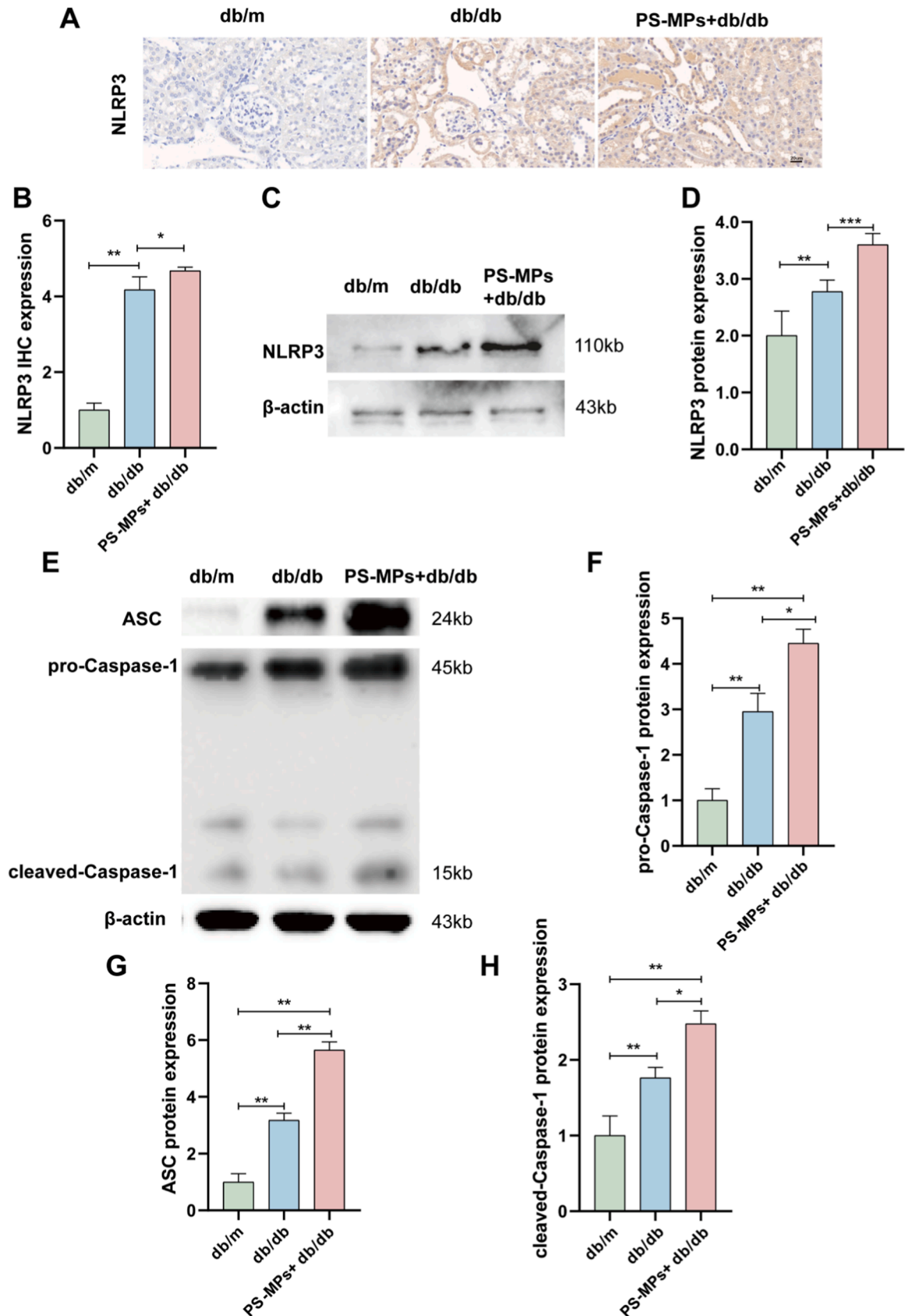


Fig. 5. PS-MPs promote the inflammasome NLRP3 in the kidneys of db/db mice. (A-B) Immunohistochemical analysis and quantification of NLRP3 protein expression. magnification, 400×, scale bar, 20 μm. (C-H) Western blot analysis and quantification of protein expressions of NLRP3, ASC, pro-Caspase-1, and cleaved Caspase-1 in different groups. β-actin was used as an internal control. Statistical analysis was performed using one-way ANOVA (A-F), n=10, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

3.7. PS-MPs aggravated renal EMT by promoting TGF- β 1/Smad2/3 signaling pathway

TGF- β plays a pivotal role in the pathogenesis of fibrosis in the majority of kidney diseases. TGF- β 1 induces renal fibrosis by activating the canonical Smad2/3 signaling pathway, ultimately leading to the activation of myofibroblasts, excessive production of ECM, and inhibition of ECM degradation. This process contributes to the development of fibrosis, which is a common feature of chronic kidney diseases. Wang et al. demonstrated that PS-MPs are capable of activating the ROS/TGF- β /Smad2/3 signaling pathway, ultimately inducing liver fibrosis injury (Wang et al., 2023). Furthermore, Zou et al. found that 5 μ m MPs promote the expression levels of TGF- β 1 and various fibrotic markers in the renal tissue of mice (Zou et al., 2022). Recent research demonstrated the significant involvement of the TGF- β 1/Smad2/3 signaling pathway in the pathogenesis of renal fibrosis (Lan, 2011). Acting as a receptor complex formed by TGF- β ligand binding, this pathway triggers the

phosphorylation of Smad proteins, thereby promoting their nuclear translocation. Once inside the nucleus, they collaboratively modulate the transcriptional activities of target genes, contributing to the development and progression of renal fibrosis. Consequently, it was attempted to investigate the potential involvement of the TGF- β /Smad2/3 signaling pathway in PS-MPs-induced renal fibrosis, and a series of pertinent tests were conducted to explore this hypothesis. The expression levels of p-Smad2 and p-Smad3 in the kidney tissue of untreated db/db mice were significantly higher than those in the control group. In addition, IHC results confirmed that PS-MPs could promote p-Smad2 protein expression level in kidney tissues of db/db mice (Fig. 6A and B). Additionally, PS-MPs significantly increased the expression levels of these proteins (Fig. 6C-E). Consequently, the findings indicated that PS-MPs could exacerbate renal fibrosis by promoting TGF- β 1/Smad2/3 signaling pathway. Activation of the TGF- β 1/Smad signaling pathway could trigger renal fibrosis, resulting in myofibroblast activation, excessive production of ECM, and inhibition of ECM degradation. This

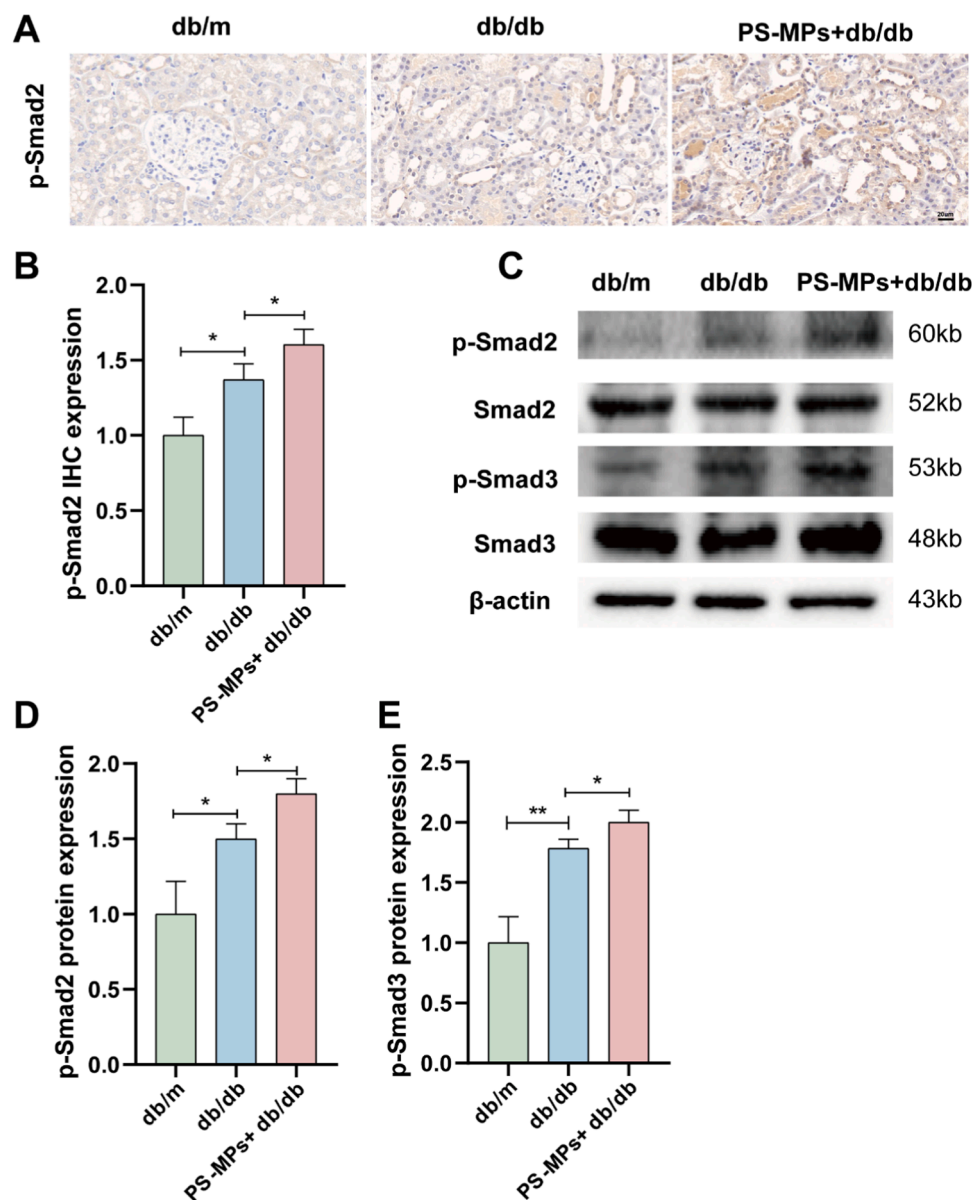


Fig. 6. PS-MPs promote TGF- β 1/Smad2/3 signaling pathway aggravating renal fibrosis in db/db mice. (A-B) Immunohistochemical staining and quantitative analysis of p-Smad2 in mouse kidney tissue. magnification, 400 $\times$, scale bar, 20 μ m. (C-E) The protein expression of p-Smad2 and p-Smad3 was examined by immunoblot from mouse kidney lysates. Statistical analysis was performed using one-way ANOVA (A-D), n=10, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, not significant.

study contributes to a more comprehensive comprehension of the mechanisms by which TGF- β 1 regulates fibrosis and provides valuable insights into the pathogenesis of renal fibrosis.

4. Discussion

As PS-MPs are increasingly released into the environment, their influences on humans have remarkably attracted scholars' attention. Multiple studies have shown that PS-MPs can be accumulated in organs/tissues of mice, such as viscera, liver, and kidneys (Deng et al., 2018; Yang et al., 2019). Mice exposed to PS-MPs experienced worsening of mild intestinal inflammation by modifying the gut microbiota (Li et al., 2020). There are numerous studies on the effects of PS-MPs on the liver, such as lipid accumulation, apparent changes in liver histopathology, etc (Lu et al., 2018; Luo et al., 2019). PS-MPs can be accumulated in murine kidneys. However, they did not assess the effects of PS-MPs on renal fibrosis or inflammation (Deng et al., 2017b). In the present study, the db/db mouse model was used to evaluate the effects of PS-MPs on DN *in vivo*. It was found that oral administration of PS- PS-MPs increased the expression levels of pro-inflammatory cytokines and promoted renal fibrosis progression in DN mice. Significant elevations in body weight, FBG, Scr, BUN, and BUA were noted in DN mice, aligning with previously reported findings (Zhu et al., 2020; Du et al., 2020). Studies have demonstrated that uncontrolled blood sugar plays a crucial role in renal tissue damage and functional decline in DN. Chronic hyperglycemia induces oxidative stress and increases TGF- β expression level, which is associated with ECM deposition and leads to glomerulosclerosis and tubulointerstitial fibrosis (Zhu et al., 2020; Du et al., 2020). Therefore, it is essential to find the influential factors of blood sugar to control the progression of DN. In the present study, it was revealed that PS-MPs could significantly increase blood glucose level in db/db mice, suggesting that the renal effects of PS-MPs may be related to their increased blood glucose level.

Furthermore, histopathological results revealed inflammatory cell infiltration and collagen deposition in the kidneys of untreated DN mice. Exposure to PS-MPs aggravated hematological parameters and renal pathological changes in DN mice. These results suggested that PS-MPs could be an essential factor for preventing and alleviating renal injury and renal fibrosis.

Studies have pointed out that hyperglycemia increases the release of pro-inflammatory cytokines, leading to renal inflammation and fibrosis (Wu et al., 2020). Several pro-inflammatory cytokines, such as TNF- α , IL-6, IL-1 β , and IL-18, have been reported to be involved in the development of DN (Wu et al., 2020). In the present study, the levels of TNF- α , IL-6, IL-1 β , and MCP-1 were elevated in the kidney tissue of db/db mice, which was consistent with previously reported findings (Wu et al., 2020; Lu et al., 2020). Exposure to PS-MPs promoted this development, suggesting their promoting effect on renal inflammation. The TGF- β 1/Smad signaling pathway is a classic fibrotic signaling pathway. TGF- β 1 can activate Smad2/3 through phosphorylation to accelerate the fibrosis process, which is also a critical pathway for the formation of EMT (Jin et al., 2019). The increased expression levels of α -SMA and FN are the main feature of EMT. In the present study, the expression levels of TGF- β 1, p-Smad2/Smad2, and p-Smad3/Smad3 were significantly upregulated in kidney tissues of untreated db/db mice, and this effect was significantly increased after exposure to PS-MPs. Furthermore, the results of Western blotting and IHC indicated that PS-MPs significantly increased the expression levels of FN, α -SMA, and COL-1 compared with those in untreated db/db mice. The findings indicated that PS-MPs could induce EMT production, possibly through TGF- β 1.

Numerous studies have demonstrated that the NLRP3 inflammasome, which is comprised of NLRP3 protein, Caspase-1, and ASC, plays a crucial role in DN kidney inflammation and primarily functions via the expression of podocytes, endothelial cells, and renal tubular epithelial cells (Moreels et al., 2008; Shintani, 2009). Once cells are stimulated, NLRP3 swiftly activates and recruits ASC and Caspase-1, thereby

promoting the activation of Caspase-1 and resulting in the secretion of pro-inflammatory cytokines, such as IL-1 and IL-18 (Zhang and Wang, 2019). Prior research demonstrated that inflammation and cell damage caused by NLRP3 are closely associated with the progression of DN, exacerbating renal fibrosis, glomerulosclerosis, and renal tubular damage (Lin et al., 2020). In the present study, the results of Western blotting and IHC indicated that the expression levels of NLRP3, ASC, Caspase-1, and IL-1 β were significantly elevated in the kidneys of DN mice. In addition, abnormal glomerular structure and massive inflammatory cell infiltration and fibrosis were also observed in the DN group. However, these phenomena were significantly exacerbated in DN mice exposed to PS-MPs. A previous study has reported that NLRP3 deficiency could reduce renal inflammation and fibrosis in DN mice (Wu et al., 2018). Inhibition of NLRP3 inflammasome activation by silencing NLRP3/ASC or inhibiting Caspase-1 activity has been shown to reduce IL1 β production and alleviate renal injury (Liu et al., 2018).

In the present study, the inflammatory and fibrotic effects of PS-MPs on DN were examined. The findings suggested the existence of additional molecular mechanisms, including autophagy and apoptosis, might contribute to the pathogenesis of this disease. Moreover, the complex interplay of multiple interconnected pathways during disease progression underscores the need for further investigations to unravel the comprehensive role of MPs in the development of DN. Additionally, conducting clinical follow-up studies on DN patients is crucial to assess the potential risks associated with exposure to MPs in this population. These insights will not only deepen our understanding of the disease, but also promote the development of preventive and therapeutic strategies.

In conclusion, the findings of this study provided compelling evidence regarding the detrimental effects of PS-MPs on renal function. Moreover, the observed exacerbation of hyperglycemia in db/db mice following exposure to PS-MPs suggests the potential translatability of these results to human subjects, thereby underscoring the significance of glycemic control as a pivotal therapeutic objective in the management of DN (Ahmed et al., 2009). Considering the elevated risk of exposure to MPs in certain individuals, there may be a need to implement personalized treatment strategies. Additionally, unraveling the complex mechanisms involving the TGF- β 1/Smad2/3 signaling pathway and the NLRP3 inflammasome signaling pathway in clinical settings holds promise for advancing our understanding of the molecular basis governing the adverse effects of PS-MPs on human physiology. These insights provide valuable guidance for the development of targeted therapeutic interventions for this condition.

5. Conclusions

In conclusion, PS-MPs exhibited to promote the development of renal fibrosis in db/db mice. It was demonstrated that PS-MPs could aggravate inflammation in the renal tissue of DN mice by activating the NLRP3 inflammatory signaling pathway, and enhancing EMT in renal tubular epithelial cells via promoting TGF- β 1/Smad signaling pathway. Therefore, PS-MPs may be a significant potential risk factor for DN.

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CRediT authorship contribution statement

Feng Li xiang: Writing – original draft, Investigation, Data curation. **Chen Chen:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Xiong Xi:** Visualization,

Validation, Formal analysis. **Wang Xiong**: Supervision, Resources, Project administration. **Li Xin xin**: Formal analysis. **Kuang Qi hui**: Investigation. **Wei Xiao**: Validation. **Gao Li kun**: Validation, Supervision. **Niu Xuan**: Writing – original draft, Visualization, Supervision, Software, Conceptualization. **Li Qingwen**: Validation, Software, Project administration, Methodology. **Yang Jun**: Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Li Lili**: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Data curation, Conceptualization. **Lao Pengcheng**: Supervision, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2024.116102](https://doi.org/10.1016/j.ecoenv.2024.116102).

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